National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Microprocessor Programming & Interfacing	Course Code:	EE303
Program:	BS(EE)	Semester:	Fall 2017
Duration:	180 Minutes	Total Marks:	90
Paper Date:	21-12-2017	Weightage:	50%
Section:	ALL	Page(s):	2
Exam Type:	Final Exam		

Student : Name: **Roll No.** **Section:**

Instruction/Notes: It is an **OPEN BOOK/OPEN NOTES** exam, *Show complete working for each question, direct answers are not acceptable. Make assumptions if you think some data is missing, but clearly mention your assumption. Attempt each question only once.*

Question#1: [20 Marks] CLO-02

Write an assembly language program to count all non-zero bytes in the entire scratch pad of general purpose RAM (30H-7FH) of 8051 microcontroller. The counting value is stored in accumulator A, and each non-zero byte is cleared (set to value zero).

P2 P1
h2 h1
P3 h3 h4 P4

Question#2: [20 Marks] CLO– 03

Consider a disk mounted on a DC motor that has a variable speed. There are 4 small holes (h1, h2, h3 and h4) on the disk (90 degrees apart). In each revolution, the system generates four pulses (one pulse against each hole). The disk of motor is connected with 8051 such that pulse of each hole is sensed at pin P3.4 while revolving. Speed of the motor is reasonably large such that the pulses are of very short duration.

W↑ rev/s

Write an assembly language program to calculate speed of motor in RPM (revolutions per minute). Maximum value of speed can be 2400RPM. Store the lower byte of answer in R0 and higher in R1 of bank0. (*Hint:* First calculate number of revolutions per second and then find speed).

$W = n \times \frac{2\pi}{60}$

Question#3: [20 Marks] CLO – 04

Two microcontrollers (MC1 and MC2) are communicating serially at a baud rate of **9600 bps** with 8 bits data, one start bit and stop bit. Both controllers have oscillator frequency of 12MHz. MC1 initiates the communication by sending 2 bytes of data. The data is 2 digit packed BCD number stored at RAM location 30H. Data is required to be encrypted into another form before being sent. Encryption is done as follows:

1st transmit

- Unpack the 2 digit BCD number and store in R0 and R1
- Convert R0 and R1 into an alphabet. 0-9 value is converted into 'A' to 'J'

Now, send R0 and R1 to MC2. MC2 receives these bytes and store in its RAM locations 30H and 31H. This process is repeated continuously. ASCII codes of alphabets A - J are 41H - 4AH.

Examples: If MC1 has 47H stored at 30H. Data that is sent actually is 'E' and 'H'

If MC1 has 92H stored at 30H. Data that is sent is 'J' and 'C'

0	1	2	3	4	5	6	7	8	9
A	B	C	D	E	F	G	H	I	J

Question#4: [30 Marks] CLO – 05, PLO5 (SLO1-2)

You are required to design a system for controlling temperature of a room. Temperature can be detected using temperature sensors. There is a fan inside the room whose speed can be controlled by a

MC1 MC2 PTO

DC motor. This system should control speed of fan based on the temperature values. If temperature is below 35°C, fan should be running with 60% of its maximum speed. If temperature is below 25°C, the speed of fan should be further lowered. It should run with 30% of its maximum speed. In all other scenarios, fan runs at 90% speed. The fan runs with 2KHz frequency. Assume a linear relation between speed and average value of PWM.

Different components are available to you in order to design this system. *Details of the components are given below:*

Temperature sensor:

A temperature sensor is configured at a certain threshold. If temperature decreases below that threshold, sensor gives logic '0' output else it gives out logic '1'. Available temperature sensors are

1. SensorA: Threshold value is 25°C
2. SensorB: Threshold value is 50°C
3. SensorC: Threshold value is 35°C
4. SensorD:Threshold value is 30°C

DC Motor:

A DC Motor is interfaced with any microcontroller using a driver IC. Each driver IC has its own voltage specifications. Fan is connected with motor output terminals. Each motor runs with certain frequency. Available DC motors are:

1. MotorA: driver IC works at 12V, frequency = 10KHz
2. MotorB: driverIC works at 5V, frequency = 2KHz
3. MotorC: driverIC works at 5V, frequency = 4KHz

Microcontroller:

Microcontrollers available in market have different on chip modules available. Driver circuits of microcontrollers have only 24MHz oscillator. Available controllers are:

1. MC1: 8 bit controller, 2 timers, 1 external interrupt source, 1 serial module
2. MC2: 8 bit controller, 3 timers, 2 external interrupt sources, 1 serial module

Assume both microcontrollers belong to 8051 family and use same assembly.

Things to do:

1. *Identify the inputs and outputs of system [5 Marks]* T → speed
2. Chose the components according to the given scenario and make a block diagram of the system. Mention the component name you are using. [5 Marks]
3. *You are required to use on chip modules. Write complete assembly language code fulfilling all efficiency requirements. [15 Marks]*
4. *Dry run your code for two different inputs and proof correctness of your code. [5 Marks]*